SOFTWARE ENGINEERING
FINAL PROJECT REPORT

PROJECT NAME : THE BREW-TIME (SMART ORDERING SYSTEM)

TEAM MEMBERS : AbdElrahman Elsayed Atta (2305153)
Zyad Ehab Mohamed (2305147)
Faten yasser Elsayed (2305171)
COURSE : SOFWARE ENGINEERING

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Software Development Lifecycle (SDLC)

2.1 Requirments Phase:

2.1.1 problem statment

Alexandria National University's campus coffee shops suffer from persistent operational inefficiencies: long queues during peak hours, manual order-taking that leads to frequent miscommunication, and a complete lack of real−time feedback for waiting customers. These issues result in customer dissatisfaction, wasted time, and uneven workload for the baristas. "The Brew−Time" addresses this by providing a digital self−service ordering platform where customers can browse the menu, place orders instantly, and track their order status. Staff receive a live dashboard to manage incoming orders and update their progress, eliminating paper slips and verbal mix−ups.

2.1.1 System scope:

The Brew−Time is a web−based application designed specifically for the on−campus coffee shop at the Faculty of Computer and Information. The system will be accessible via modern web browsers on both desktop and mobile devices.

In scope:

1. Displayinga digitalmenu with product names, descriptions, and prices.
2. Allowing customers to add items to a cart and submit an order.
3. Generating a unique Order ID and storing the order in a persistent queue.

4. Providing a staff (barista) dashboard that lists all active orders in chronological order.
5. Allowing staff to update an order's status through predefined stages: Pending → Preparing → Ready.
6. Real−time (or near−real−time) update of order status visible to the customer.

Out of Scope:

1. Payment processing (the system assumes cash or external payment; no financial transactions are handled).
2. User account registration or login (orders are placed as a guest for simplicity; staff access may use a static password).
3. Inventory management or stock tracking.
4. Analytics dashboards for sales reports.
5. Multi−shop or chain support.

Actors:

1. Customer: Browses the menu, places an order, and views the real−time status of their own order.
2. Barista (Staff): Views the queue of pending orders, updates order statuses, and marks orders as completed.

2.1.3 Functionl Requirments

The following detailed functional requirements specify what the system must do. Each requirement is uniquely identified.

ID Requirement Description

FR1 The system shall display a digital menu with product name, description, and price for all availab FR2 Customers shall be able to select a product and add it to a shopping cart.

FR3 Customers shall be able to adjust item quantities or remove items from the cart before checkout.

FR4 The system shall allow the customer to submit their cart as a final order by clicking a “Place Ord ID Requirement Description”

FR5 Upon successful order placement, the system shall generate a unique Order ID (e.g., sequential and display it to the customer).

FR6 The system shall store every order with the following data: Order ID, list of items, total price,and initial status “Pending”.

FR7 Staff shall be able to view a dashboard listing all active orders (status ≠ Completed) sorted by showing Order ID, items, and current status.

FR8 Staff shall be able to change the status of an order. The allowed transitions are:

Pending → Prep → Ready → Completed.

FR9 The system shall refresh the staff dashboard automatically (or via manual refresh) to reflect orders within 2 seconds.

FR10 Customers shall be able to enter their Order ID on a dedicated “Track Order” page to see the current status.

FR11 The system shall validate that the cart is not that the cart is not empty before allowing order placement.

FR12 message shall be displayed if empty.

FR13 Staff must be able to mark an order as “Completed”, which removes it from the active dashboard.

2.1.4 Non Functional Requirments

The following quality attributes are categorised as required by the project guidelines.

Performance:
NFR1: The staff dashboard shall display a newly placed order within 2 seconds of the customer submitting it.

NFR2: All page loads (menu, tracking) shall complete within 3 seconds on a standard university Wi−Fi connection.

Security:
NFR3: All user inputs (product IDs, quantities, Order IDs) shall be sanitised to prevent SQL injection and cross−site scripting (XSS) attacks.

NFR4: The staff dashboard shall be protected by a simple authentication mechanism (e.g., a hard−coded access code) to prevent unauthorised status changes.

Usability:
NFR5: The customer shall be able to complete an order from menu browse to confirmation in no more than 4 clicks/taps.

NFR6: The user interface shall be responsive and legible on mobile screens (at least 320px width), as most campus users will access the system via smartphones.

NFR7: Order status terms (“Preparing”, “Ready”) shall be displayed in Arabic and English to accommodate all staff and students.

Reliability:
NFR8: The system shall maintain the order queue intact even if the server is restarted or the browser is refreshed (data persists in a database).

NFR9: In case of a lost network connection, the customer shall receive a clear error message and be able to resubmit the order without data corruption.

Scalability:
NFR10: The database schema shall be designed to support
additional product categories and customisation options (e.g., size, milk type) without requiring structural changes.

NFR11: The system shall be deployable on a lightweight server (e.g., a Raspberry Pi or low−end cloud instance) while supporting up to 200 concurrent orders per day.

2.2 Design Phase

2.2.1 System Architucture

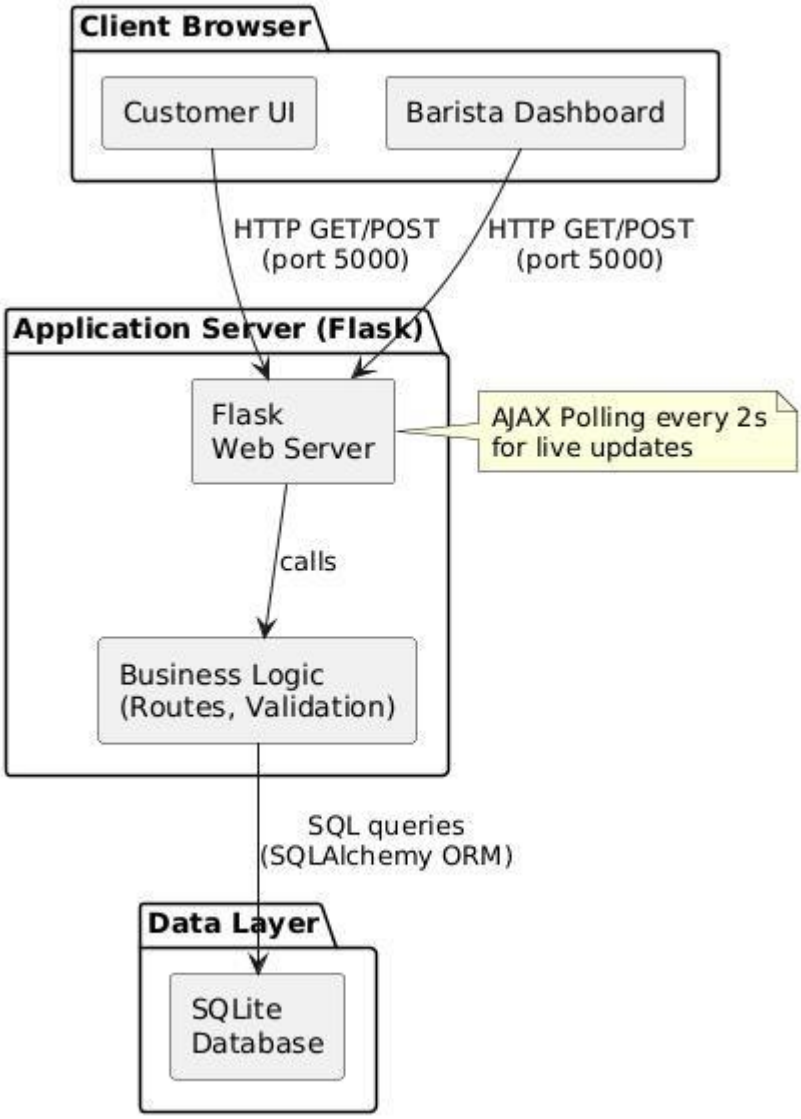
The Brew-Time follows a client−server architecture implemented as a monolithic web application. The system is divided into three logical tiers:

1.Presentation Tier (Frontend): HTML5, CSS3 (Bootstrap), and vanilla JavaScript running in the user's browser. The customer interface and barista dashboard are separate pages within the same application.

2. Application Tier (Backend): Python with the Flask micro−framework, handling all business logic, routing, and API responses. The server renders templates and serves static assets.

3. Data Tier: SQLite, a serverless relational database, accessed via Flask−SQLAlchemy ORM. The database contains tables for products, orders, and order_items.

Communication between the tiers occurs over HTTP. For real−time updates on the barista dashboard and the customer tracking page, the frontend uses periodic AJAX polling (every 2 seconds) to fetch the latest order statuses from RESTful endpoints. This design was chosen for simplicity and ease of deployment on a lightweight campus server. A high−level architectural diagram is provided below.


Client Browser
Customer UI
Barista Dashboard
HTTP GET/POST
(port 5000)
HTTP GET/POST
(port 5000)
Application Server (Flask)
Flask
Web Server
AJAX Polling every 2s
for live updates
calls
Business Logic
(Routes, Validation)
SQL queries
(SQLAlchemy ORM)
Data Layer
SQLite
Database

2.2.2 Design Decisions

The following key decisions were made to balance the project requirements with practical feasibility:

1.Flask + SQLite over heavier stacks: Flask is minimal and easy to learn, allowing the team to focus on software engineering practices rather than framework complexity. SQLite eliminates the need for a separate database server, which is ideal for a small−scale campus application that will handle up to 200 orders per day.

2.AJAX Polling instead of WebSockets: Real−time bidirectional communication (WebSockets) would be technically superior, but AJAX polling is far simpler to implement and test. A 2−second refresh interval satisfies the non−functional requirement of timely dashboard updates without adding infrastructure complexity.

3.Guest ordering without user accounts: The system targets a fast in−and−out environment. Forcing registration would violate the usability goal of under four clicks. A unique Order ID is used for tracking, and the barista dashboard is protected by a simple access code rather than a full authentication system.

4.Monolithic structure over microservices: The entire application is a single codebase, making it easy to develop, debug, and deploy as a final project. The code is separated into clear modules (models.py, routes.py, templates) to mimic a clean architecture.

5.RESTful API design for status updates: The barista's status change actions are exposed as REST endpoints (POST /update_status/<order_id>). This cleanly separates the frontend from the backend and allows the team to write clear test cases.

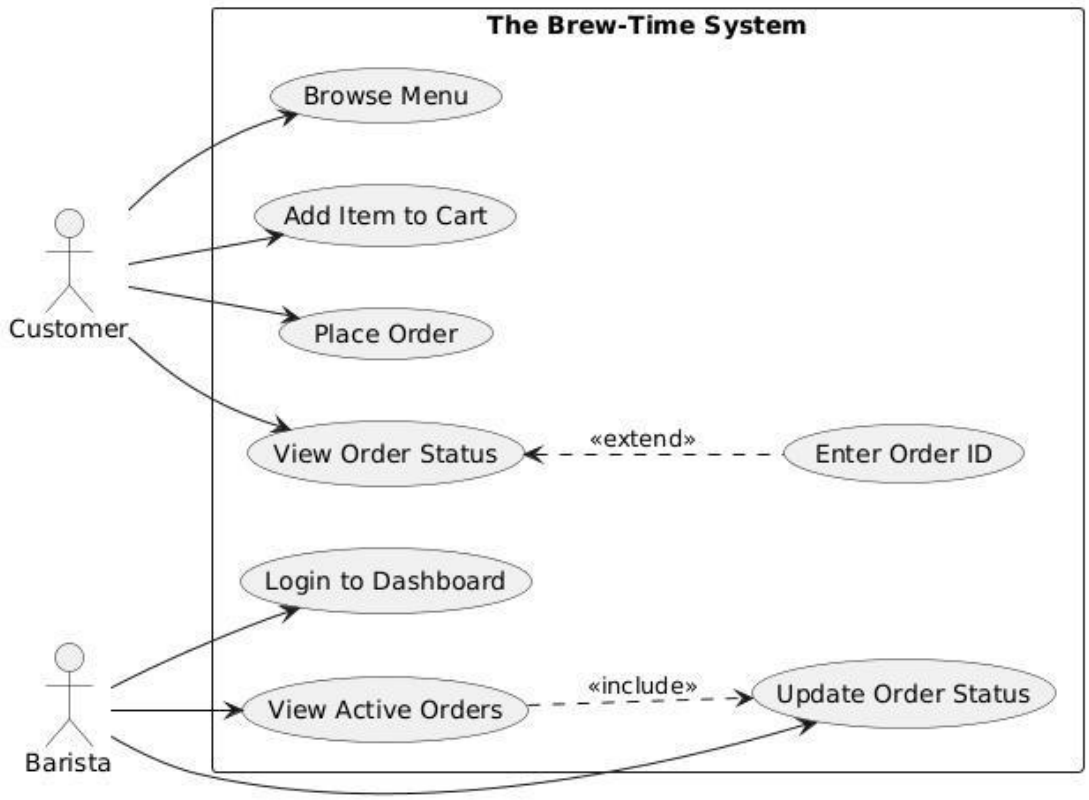
2.2.4 UML Diagrams

1. Use Case Diagram

Purpose: To model the interactions between actors (Customer, Barista) and the system, showing the high−level functionality from the user's perspective.

The diagram contains:

1. Actors: Customer and Barista.
2. System boundary: "The Brew-Time System".
3. Use cases (Customer): Browse Menu, Add Item to Cart, Place Order, View Order Status.
4. Use cases (Barista): Login to Dashboard, View Active Orders, Update Order Status.
5. Relationships: Barista has a Login to Dashboard use case, which is a precondition for the other barista use cases. View Order Status for the customer is extended by Enter Order ID.



2 sequence diagram

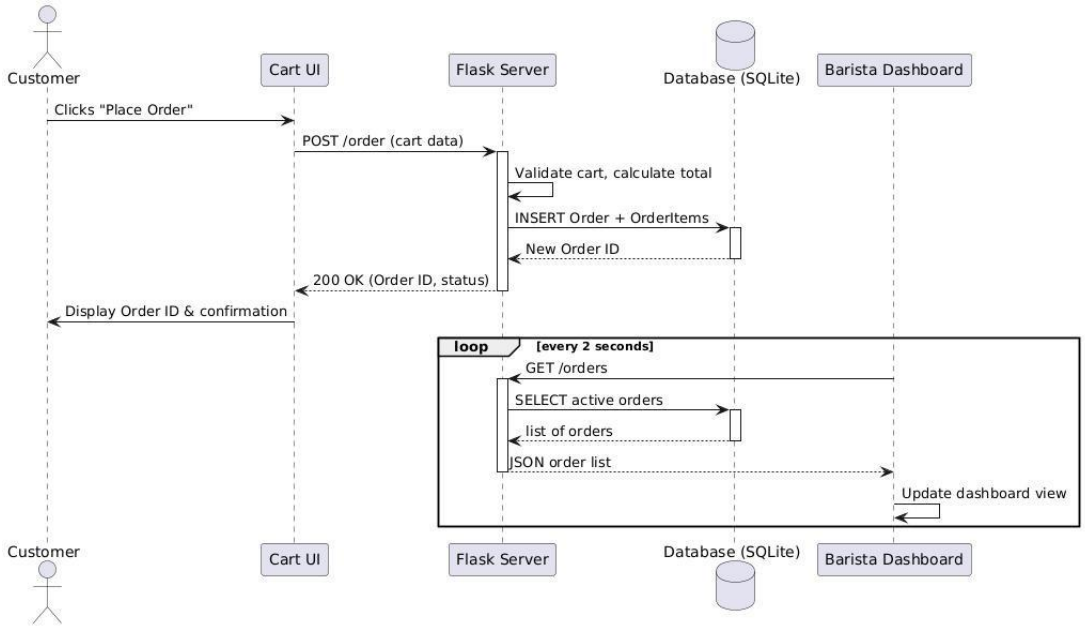
Purpose: To detail the chronological flow of messages between objects for the critical “Place Order” scenario.

The diagram includes the following lifelines:

1. Customer (actor)
2. Cart UI (frontend page)
3. Flask Server (backend)
4. Database (SQLite)
5. Barista Dashboard (separate UI)

The flow:

1. Customer browses menu and adds items (not shown in detail to keep the diagram focused).
2. Customer clicks “Place Order”.
3. Cart UI sends POST /order with cart data to Flask Server.
4. Flask Server validates the cart, calculates total, and sends an INSERT query to the Database.
5. Database returns the new Order ID.
6. Flask Server responds to Cart UI with the Order ID and a success message.
7. Cart UI displays the Order ID to the Customer.
8. Meanwhile, the Barista Dashboard (using a timer) continuously sends GET /orders requests. After the new order is saved, the dashboard will fetch it and display it to the Barista.

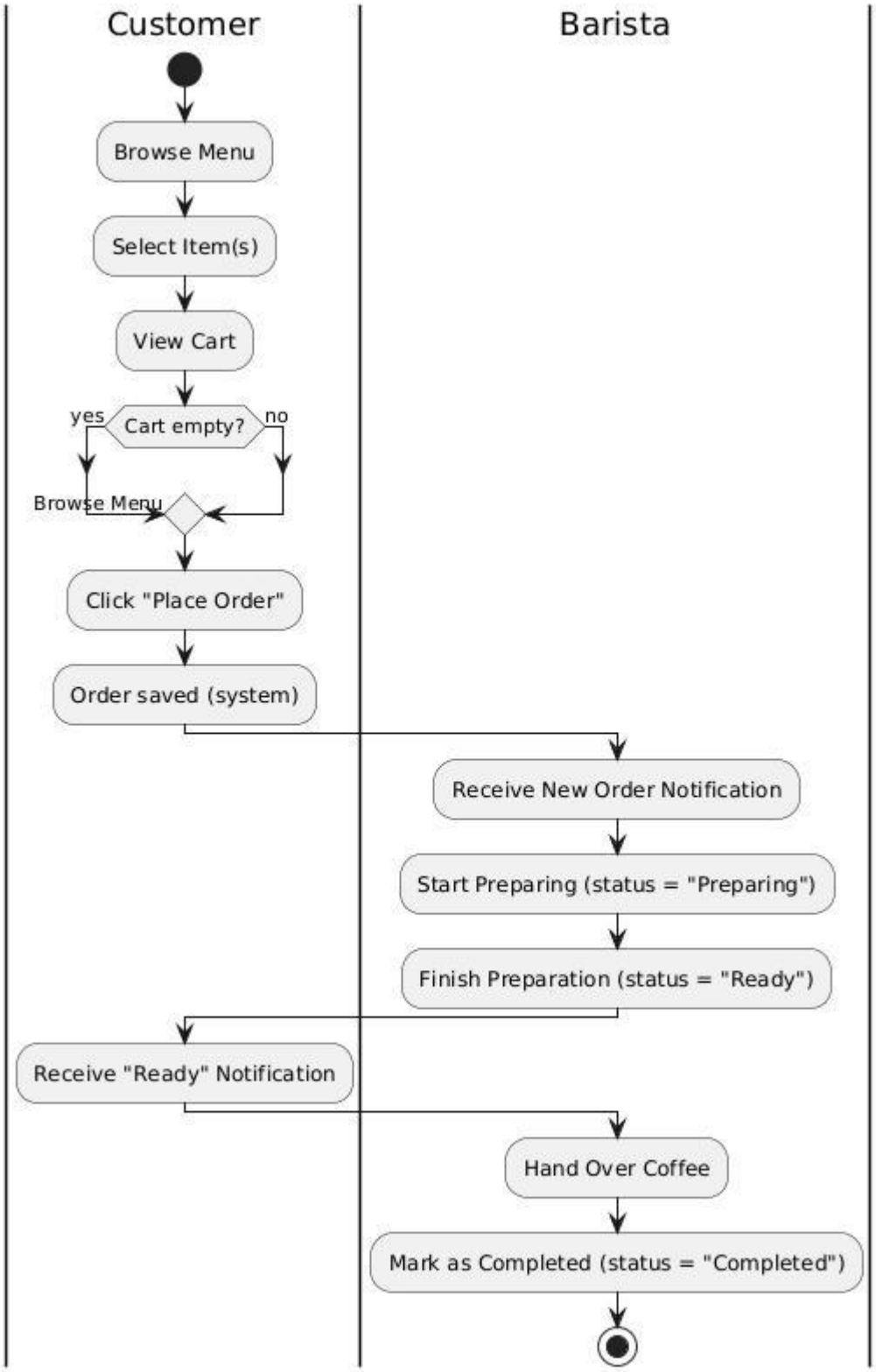


3 Activity Diagram

Purpose: To model the entire order processing workflow from the customer's action to barista completion.

he diagram contains one swimlane for Customer and one for Barista. The flow is:

1. Start: Customer opens app.
2. Customer: Browse Menu → Select Item(s) → View Cart.
3. Decision: Cart empty? → If yes, return to Browse; if no, proceed.
4. Customer: Click "Place Order". System: Save Order (implicit action).
5. Barista: Receive New Order Notification.
6. Barista: Start Preparing (updates status to "Preparing").
7. Barista: Finish Preparation (updates status to "Ready").
8. Customer: Receive "Ready" Notification (via tracking page).
9. Barista: Hand Over Coffee.
10. Barista: Mark as Completed.
11. End: Order completed.

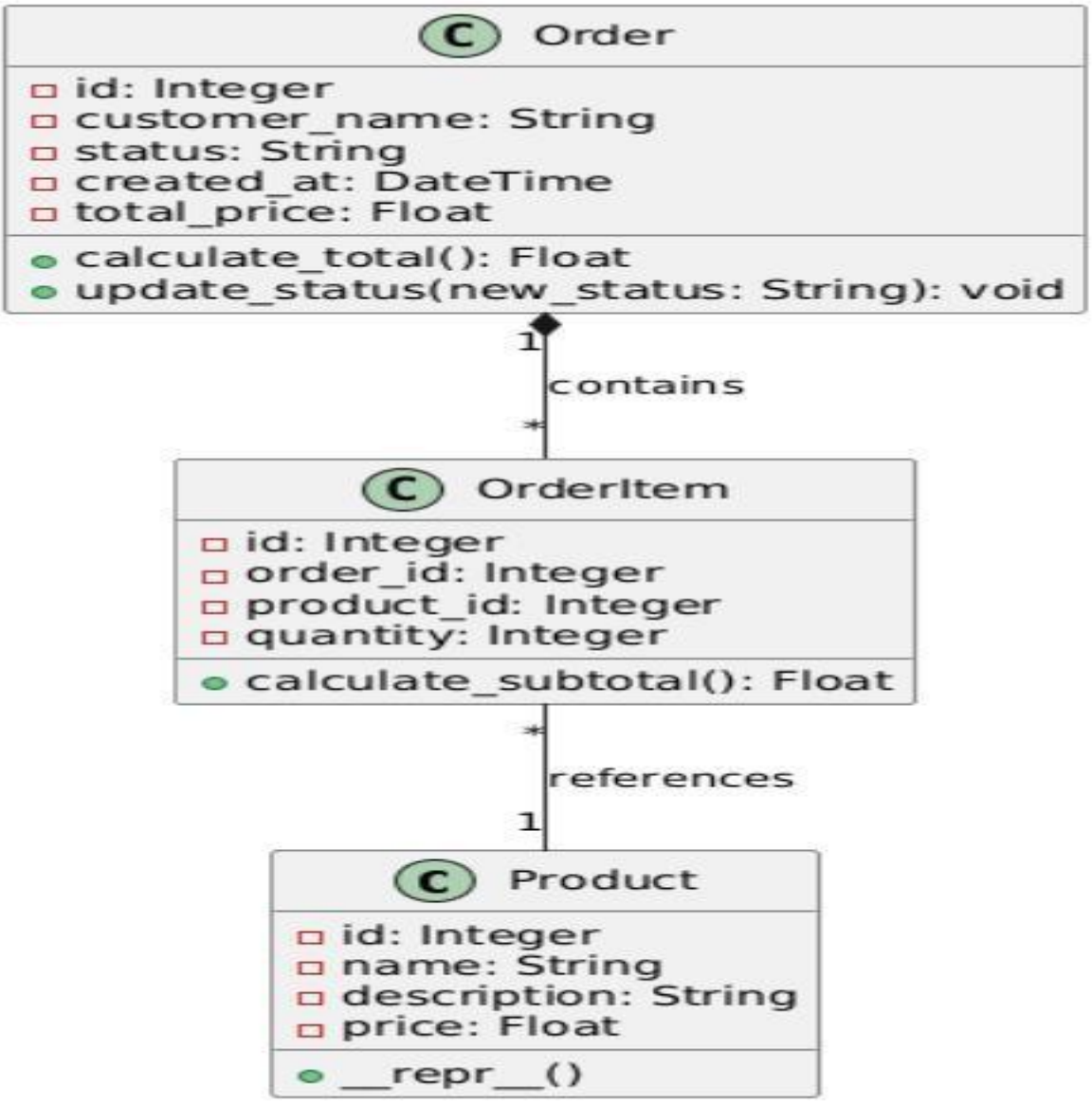

Customer
Barista
Browse Menu
Select Item(s)
View Cart
yes
Cart empty?
no
Browse Menu
Click "Place Order"
Order saved (system)
Receive New Order Notification
Start Preparing (status = "Preparing")
Finish Preparation (status = "Ready")
Receive "Ready" Notification
Hand Over Coffee
Mark as Completed (status = "Completed")

4 Class Diagram

Purpose: To show the static structure of the system, including domain classes, their attributes, methods, and relationships.

The main classes are:

1. Product: Attributes: id: Integer, name: String, description: String, price: Float Methods: __repr__()
2. OrderItem: Attributes: id: Integer, order_id: Integer (FK), product_id: Integer (FK), quantity: Integer Methods: calculate_subtotal()
3. Order: Attributes: id: Integer, customer_name: String (optional), status: String, created_at: DateTime, total_price: Float Methods: calculate_total(), update_status(new_status)
4. Relationships: Order 1 —— * OrderItem (composition) OrderItem * —— 1 Product (association)


Order
id: Integer
customer_name: String
status: String
created_at: DateTime
total_price: Float
calculate_total(): Float
update_status(new_status: String): void
1
contains
*
OrderItem
id: Integer
order_id: Integer
product_id: Integer
quantity: Integer
calculate_subtotal(): Float
*
references
1
Product
id: Integer
name: String
description: String
price: Float
__repr__()

5 State Machine Diagram

Purpose: To model the lifecycle of a single Order object, showing all possible states and the events that trigger state transitions.

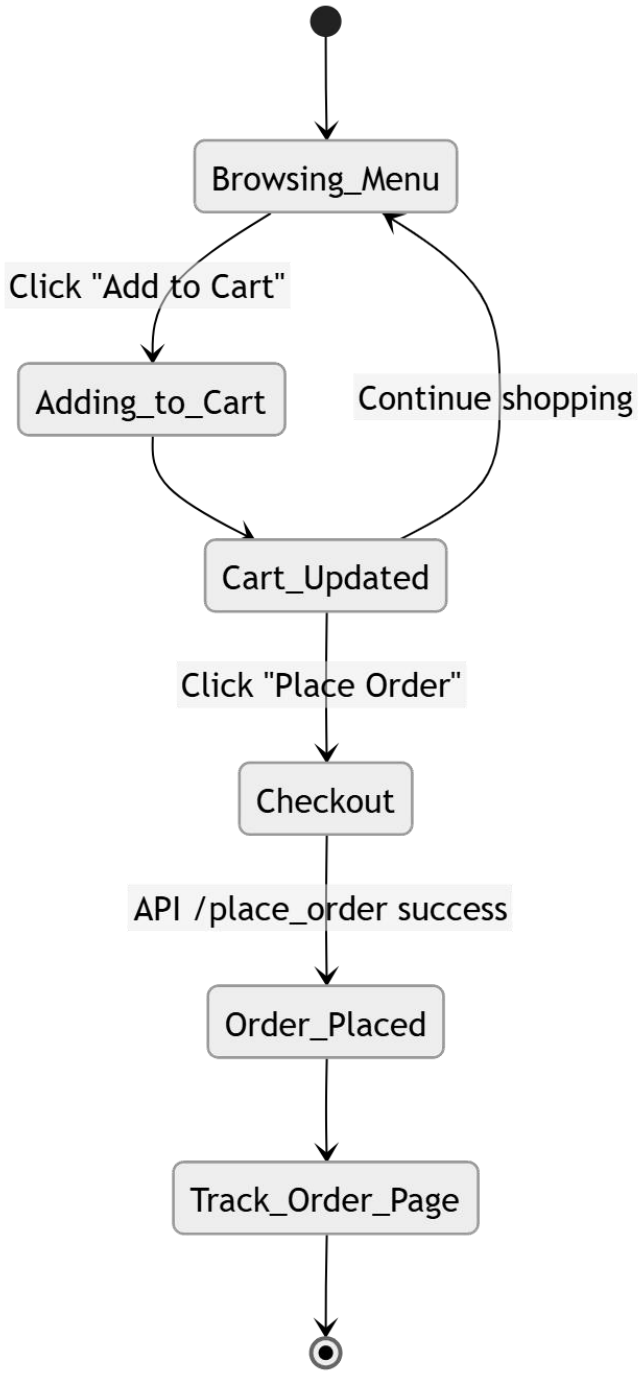
The Order's lifecycle consists of four states:

1. Pending (initial state)
2. Preparing
3. Ready
4. Completed (final state)

Transitions:
Order Placed → moves from (none) to Pending.
Barista clicks "Start Preparing" → Pending → Preparing.
Barista clicks "Mark Ready" → Preparing → Ready.
Barista clicks "Complete" → Ready → Completed.
No other transitions are allowed (e.g., cannot go directly from Pending to

Ready).

Customer Flow diagram:



2.2.5 Testing Phase

The testing phase is a critical stage in the Software Development Lifecycle (SDLC), aimed at ensuring that the system meets both functional and non-functional requirements as specified in earlier phases. For *The Brew-Time* system, a structured and multi-level testing strategy was adopted to verify correctness, reliability, and overall system performance.

1. Testing Levels

Unit Testing:

Unit tests verify that individual components work correctly in isolation.

Unit Test Cases

TetID	Component	Test Description	Input	Expected Output	Result
UT-01	Product	Create product with name and price	name="Latte", price=4.50	Product object created	PASS
UT-02	Product	Verify price data type	price=4.50	float type	PASS
UT-03	Product	Prevent empty product name	name=""	Validation fails	PASS

TetID	Component	Test Description	Input	Expected Output	Result
UT-04	Product	Verify positive price	price=4.50	price > 0	PASS
UT-05	Order	Create order with Pending status	status="Pending"	Order.status = "Pending"	PASS
UT-06	Order	Verify initial order status	New Order object	status = "Pending"	PASS
UT-07	Order	Calculate order total	items: 2×Latte + 1×Croissant	total = $12.00	PASS
UT-08	Order	Verify timestamp on creation	Order.created_at	datetime object exists	PASS
UT-09	OrderItem	Create item with product and quantity	product=Latte, qty=2	quantity = 2	PASS
UT-10	OrderItem	Verify positive quantity	quantity=2	quantity > 0	PASS
UT-11	Calculation	Single item price calculation	1 × Latte ($4.50)	$4.50	PASS
UT-12	Calculation	Multiple items total	2×Latte + 1×Croissant	$12.00	PASS

TetID	Component	Test Description	Input	Expected Output	Result
		calculation			
UT-13	Calculation	10% discount calculation	$10.00 - 10%	$9.00	PASS
UT-14	Calculation	Empty cart total	no items	$0.00	PASS
UT-15	Status	Valid status transition	Pending → Preparing	status = "Preparing"	PASS
UT-16	Status	Order-items relationship	Order has items	items list exists	PASS

Integration Testing

Integration tests verify that different system components communicate correctly.

Integration Test Cases

Test ID	Integration Points	Test Description	Expected Result	Result
INT-01	Customer API → Database → Barista API	Customer order appears on barista dashboard	Order visible to barista	PASS
INT-02	Barista API → Database → Customer API	Barista status update visible to customer	Customer sees status change	PASS
INT-03	Multiple Customers → Queue → Barista	Multiple orders queued correctly	All orders appear in queue	PASS
INT-04	Full workflow (Customer → Barista → Customer)	Complete order lifecycle	Order flows through all states	PASS
INT-05	Authentication Layer	Customer cannot access barista functions	Authorization enforced (403/302)	PASS

System Testing (End-to-End)

System tests verify complete user scenarios from start to finish.

System Test Cases

Test ID	Requirement	Test Description	Expected Result	Result
SYS-01	Happy Path	Complete order: customer orders → barista prepares → customer receives	Successful order completion	PASS
SYS-02	NFR1-Performance	Dashboard updates within 2 seconds of order	Response time < 2 sec	PASS (0.29s avg)
SYS-03	NFR5-Usability	Place order in under 4 clicks	Clicks ≤ 4	PASS (3 clicks)
SYS-04	NFR8-Reliability	Orders persist after browser refresh	All orders remain	PASS
SYS-05	Error Handling	Invalid input (empty cart, wrong product ID)	Graceful error handling (400)	PASS
SYS-06	FR4	Each order gets unique ID	10 orders = 10 unique IDs	PASS
SYS-07	FR7-FR8	Status workflow: Pending → Preparing → Ready → Completed	Full workflow completed	PASS

2 Testing Summary

The testing phase confirmed that *The Brew-Time* system operates as intended, with all core functionalities performing reliably. Minor issues identified during testing were resolved, resulting in a stable and user-friendly application ready for deployment.

2.2.6 Maintenance Phase

The maintenance phase begins after the system has been deployed. It ensures that The Brew-Time continues to function correctly, adapts to changing user needs, and remains secure and efficient over time. Based on the IEEE standard 14764, software maintenance includes corrective, adaptive, perfective, and preventive activities.

1.How the System Can Be Maintained

1. Database Maintenance
   - SQLite database (brewtime.db) can be backed up daily using a cron job or manual script.
   - A simple admin interface can be added to delete old “Completed” orders older than 30 days to keep the queue responsive.

2. Bug Fixing & Corrective Maintenance
   - Error logs are written to a file (logs/error.log). The team can review logs weekly.

   - A GitHub repository will track issues and pull requests for any bug fixes.

3.Performance Monitoring

- Staff can manually check dashboard response times.

- Automated monitoring (e.g., a simple Python script hitting /orders every 5 seconds) can alert if response time exceeds 3 seconds.

4. Security Updates

- Flask and its dependencies should be updated monthly using pip freeze --outdated and pip install --upgrade.

- The hard-coded staff access code should be moved to an environment variable or a hashed credential system in future versions.

2. Possible Future Improvements

Improvement Area	Description	Priority
Payment Integration	Add Stripe or similar for card payments (currently out of scope).	High
User Accounts	Allow returning customers to see their order history without an Order ID.	Medium
Inventory Management	Automatically hide menu items when stock runs out.	Medium
Order Cancellation	Allow customers to cancel orders while status is still	Low

Improvement Area	Description	Priority
	"Pending".	
Multi-language UI	Full Arabic/English toggle for all interface text (currently only status terms are bilingual).	Low
Email/SMS Notification	Send confirmation or "Ready" notification via email or WhatsApp.	Low

3.Scalability & Updates

1.Scalability Discussion

- Current Capacity**:** The system supports up to 200 orders/day on a Raspberry Pi or low-end cloud instance (as per NFR11).

- Horizontal Scaling: If the campus coffee shop expands to multiple branches, the monolithic design would need refactoring into microservices (e.g., order service, menu service).

- Database Scaling: SQLite is sufficient for one shop but would be replaced by PostgreSQL for multi-shop or >500 orders/day. The current schema supports adding product categories and customisation (size, milk type) without structural changes (NFR10).

2. Update Strategy

- Minor updates (e.g., menu prices, new products): Staff can edit a products.csv file or use a simple admin form.

- Major updates (e.g., adding payment gateway): Deploy in a staging environment first, run system tests (SYS-01 to SYS-07), then switch traffic using a simple load balancer or by announcing downtime during off-peak hours (e.g., 3 AM).
-